Does Difficulty-Aware Curriculum Learning Help RL Fine-Tuning of LLMs?

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CS 224R: Reinforcement Learning · Stanford University

Problem & Motivation

We fine-tune **Qwen2.5-0.5B** on the **Countdown** arithmetic reasoning task (given a set of numbers, build an expression hitting a target) through the standard stack: **SFT** warm-start → preference optimization (**IPO**) → online policy gradient (**RLOO**).

Research Question

RL post-training drives recent reasoning gains, and *curriculum learning* is widely assumed to help by exposing the model to problems from easy to hard and avoiding wasted effort on too-easy / too-hard examples [1, 2]. **But does it actually help across SFT, IPO, and RLOO?** And can a *cheap, exact* difficulty signal drive it?

- **Why it matters:** sparse rewards and limited compute make *which* problems you train on, and *in what order*, a natural lever, but controlled evidence in LLM fine-tuning is thin.
- We test curricula **head-to-head against uniform sampling** in one controlled setting, with a difficulty we can compute exactly.

Our extension is difficulty-aware curriculum learning and difficulty-weighted objectives across SFT, IPO, and RLOO.

Method I: An Exact Difficulty Signal

For every Countdown instance we run dynamic programming over all sub-expressions to compute a pretraining difficulty signal that is deterministic and cheap relative to model training:

- **#solutions**: distinct expressions reaching the target,
- **requires** $\times, \div$: no $+/-$ only solution exists,
- **#operands** (3 or 4 numbers).

A score $\log_{\text{inv}} = -\log(\#\text{sol}+1)$ (higher = harder) is used weight the importance of each prompt.

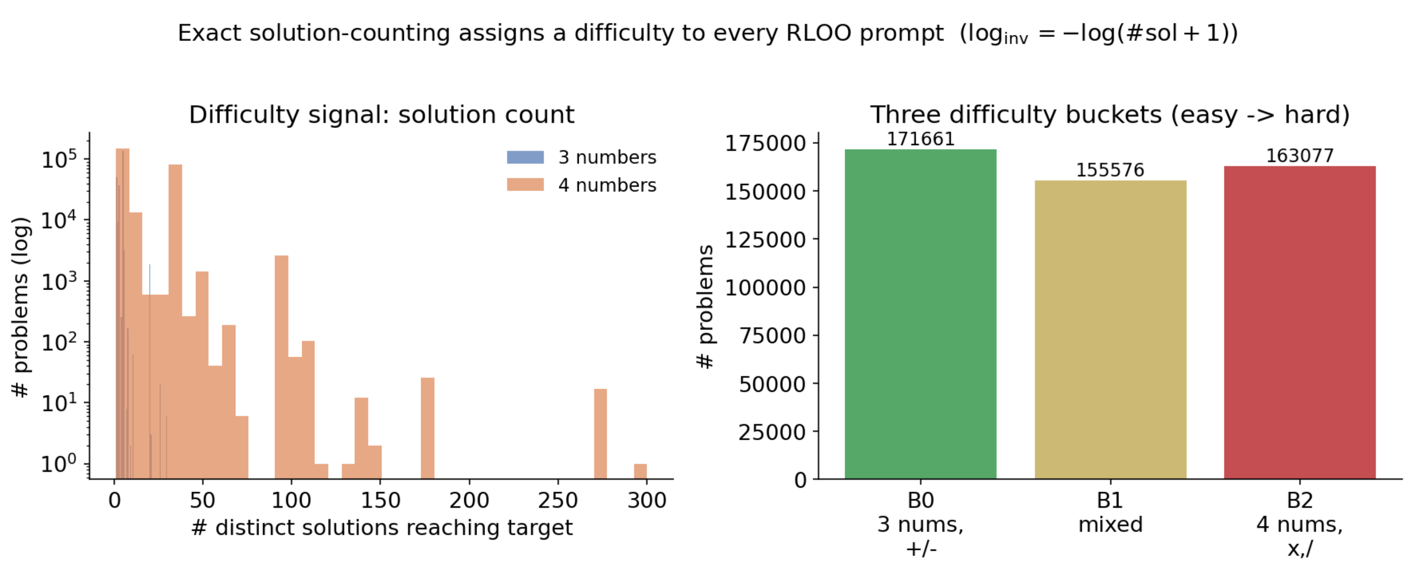

Figure 1. Exact solution-counting yields a per-problem difficulty and a three-tier bucketing of the RLOO prompt set (~490k prompts, well balanced across buckets).

Method II: Curricula & Weighted Loss

We test two uses of difficulty: changing the prompt distribution and changing the per-example loss weight. A single difficulty-bucketed sampler, shared across all three trainers:

- **Fixed (a priori):** sampling distribution slides easy → hard over training (Gaussian over buckets, center $0 \to 2$).
- **Adaptive (competence):** sample bucket $b \propto p_b(1-p_b)$, where p_b is an EMA of measured per-bucket accuracy, focuses on the $p \approx 0.5$ *edge of learnability* [2].
- **Difficulty-weighted loss:** per-prompt weight $w \propto \exp(\gamma\, z(\log_{\text{inv}}))$ up-weights harder prompts. This changes the *objective* $\sum_i w_i \ell_i$, not just the visitation order. Here z: standardized $\log_{\text{inv}}$ and γ: weighting strength

Experimental Setup

- **Model:** Qwen2.5-0.5B (base) + SFT countdown checkpoint for IPO / RLOO.
- **Eval:** 50 held-out prompts × 16 samples (vLLM, $T{=}0.6$, top-p 0.95). A response is *correct* only if the verifier confirms an exact, valid equation.
- **Metric:** **pass@k**, unbiased HumanEval estimator
- **Comparison:** uniform sampling (**Default**) vs. **Fixed**, **Adaptive**, and difficulty-**weighted** loss.

Headline Result: pass@1

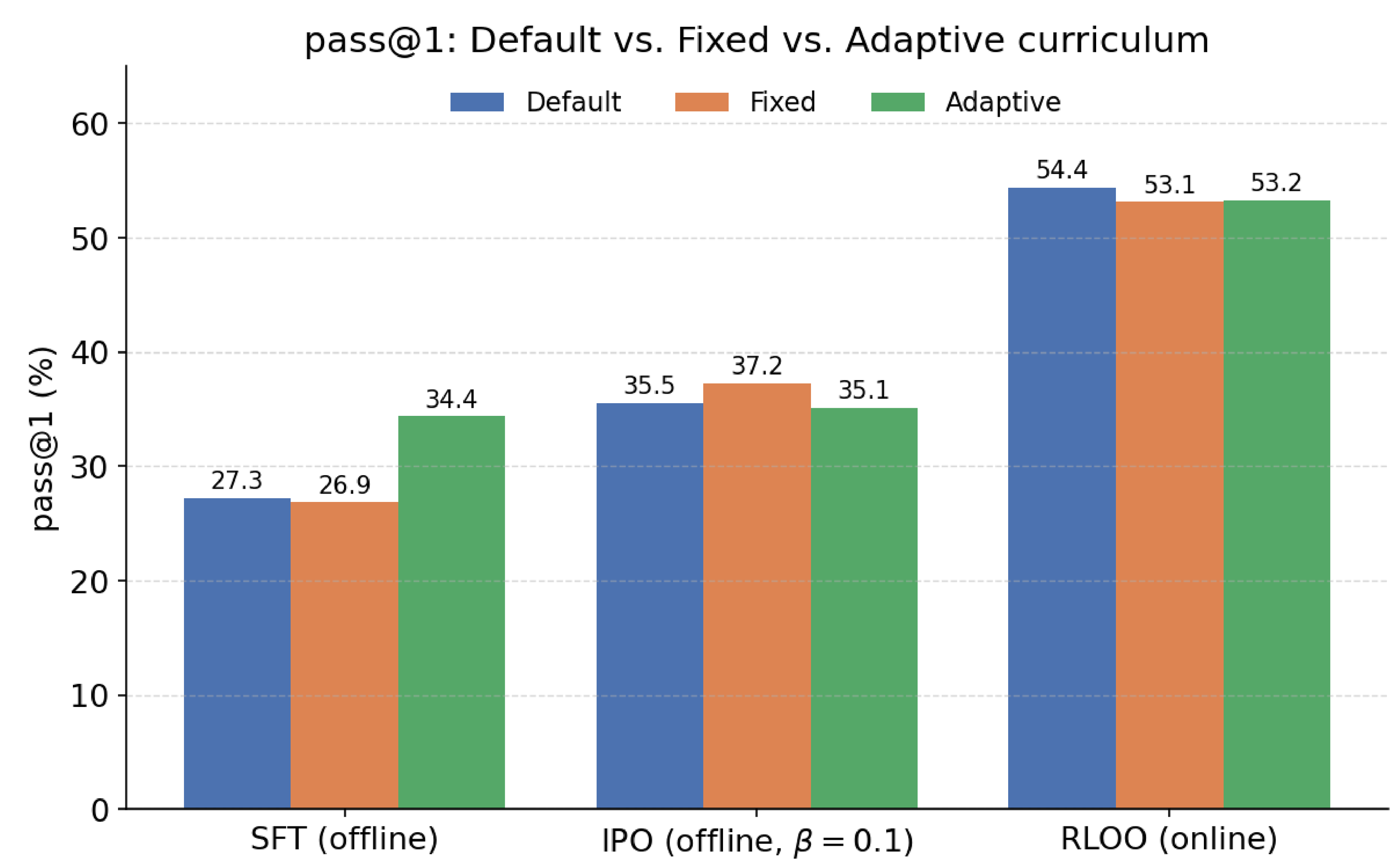

Figure 2. Headline pass@1 comparison across SFT, IPO, and RLOO. Curricula mostly match uniform sampling; adaptive SFT is the only clear pass@1 gain.

Method	SFT	IPO	RLOO
Default (uniform)	27.3 / 74	35.5 / 80	54.4 / 76
Fixed curriculum	26.9 / 78	**37.2** / 76	53.1 / 74
Adaptive curriculum	**34.4** / 78	35.1 / 78	53.2 / 74

Table 1. pass@1 / pass@16 (%). Curricula match Default within noise *except* SFT-Adaptive (**+7.1** pass@1).

Pass@k Curves

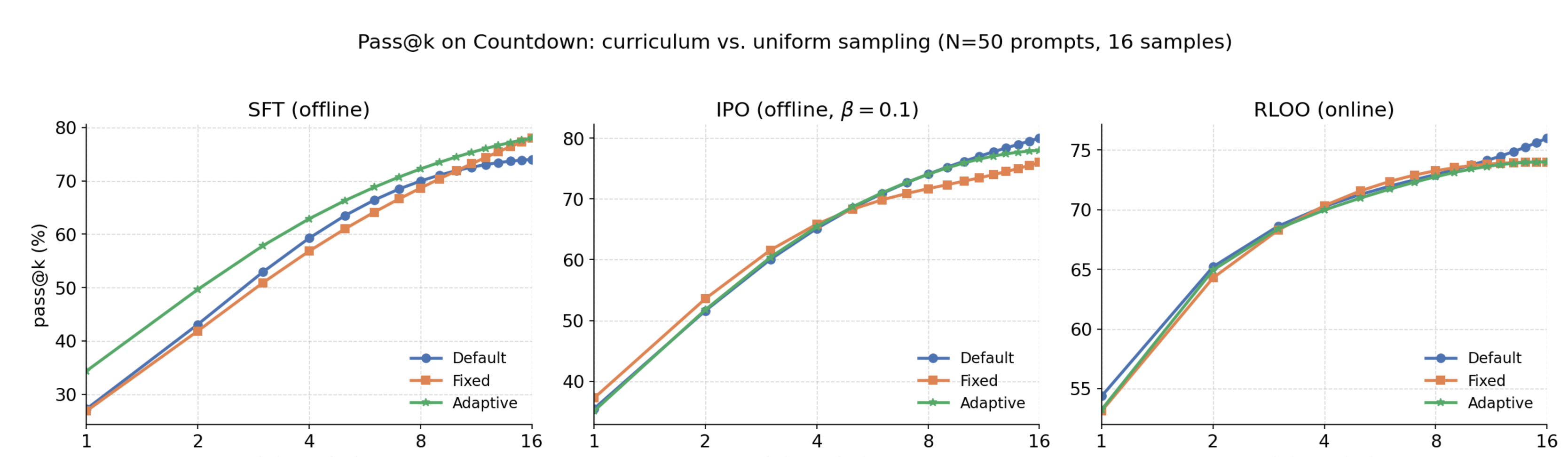

Figure 3. Curves overlap heavily. **pass@16 stays ≈74–80% everywhere**, curricula reshape pass@1 *sharpness*, not the set of solvable problems.

Where Does It Help?

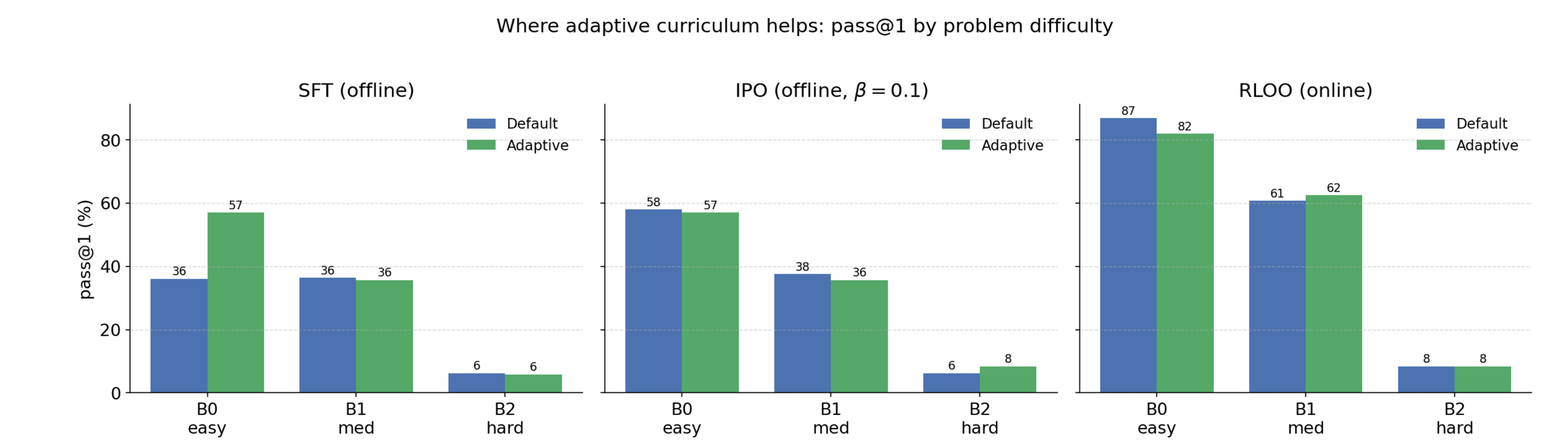

Figure 4. pass@1 by difficulty bucket. Adaptive SFT gains come from easy B0 prompts, not the hard B2 tail.

Breaking pass@1 down by problem difficulty:

- SFT-Adaptive's gain is **concentrated in easy (B0) problems** ($36 \to 57\%$), not the hard tail.
- The **hard bucket (B2) stays near 0** for *every* method $\implies$ curriculum does not unlock genuinely hard problems at pass@1.
- IPO and RLOO are tied with Default across all buckets.

Difficulty-Weighted Loss

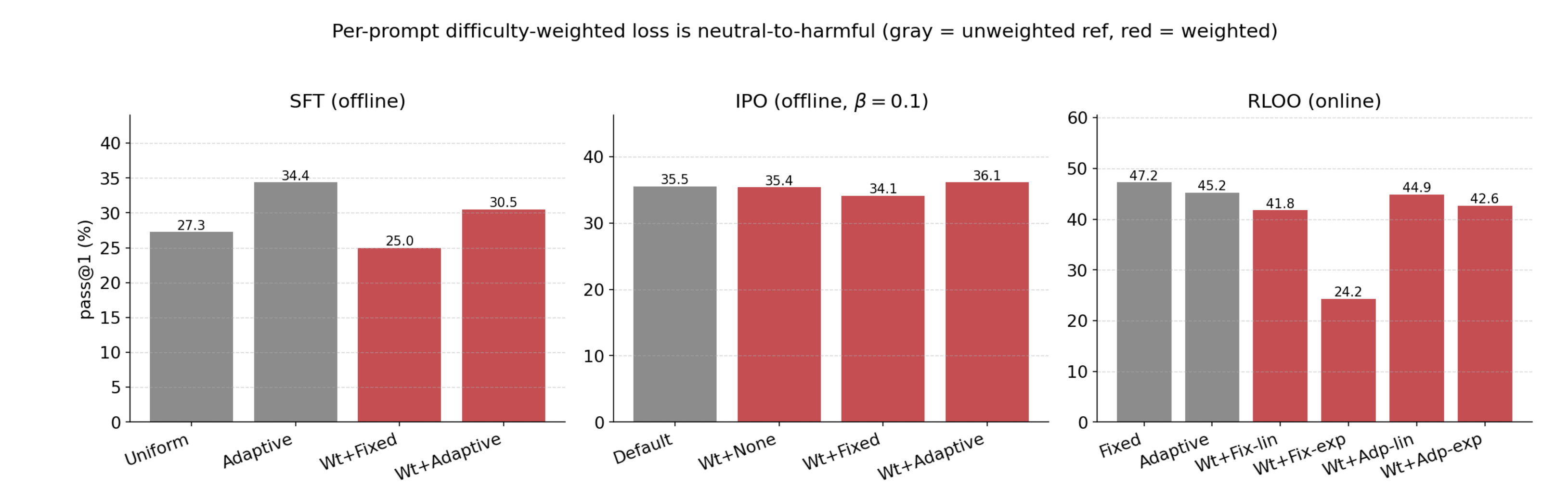

Figure 5. Neutral-to-harmful across all three trainers. **RLOO is most fragile**: exponential weighting collapses the fixed curriculum ($47 \to 24$ pass@1); only adaptive+linear stays safe (pass@1 flat, pass@16 $76 \to 82$).

Key Findings & Takeaways

1. **Curriculum gains are small and paradigm-specific.** Only *adaptive* (competence-based) curriculum helps, and only for SFT; *fixed* schedules never beat uniform.
2. **Why?** Offline objectives (SFT/IPO) over ~1 epoch are largely order-invariant; RLOO's leave-one-out advantage is *zero* whenever a prompt's rollouts share a reward, so re-weighting cannot create learning signal.
3. An a priori difficulty signal is great for *analysis* but a weak *training* lever.

Limitations & Future Work

Single seed, 50-prompt eval (noisy), 0.5B model, coarse 3-bucket difficulty. **Next:** use #solutions to *select RLOO rollouts* (prioritize mixed-success prompts that give non-zero advantage) rather than weighting the loss; finer continuous difficulty buckets; use negative log inverse to create difficulty buckets.

References

[1] S. Parashar et al. Curriculum reinforcement learning from easy to hard tasks improves llm reasoning. *arXiv:2506.06632*, 2025.

[2] S. Sundaram et al. Teaching models to teach themselves: Reasoning at the edge of learnability. *arXiv:2601.18778*, 2026.